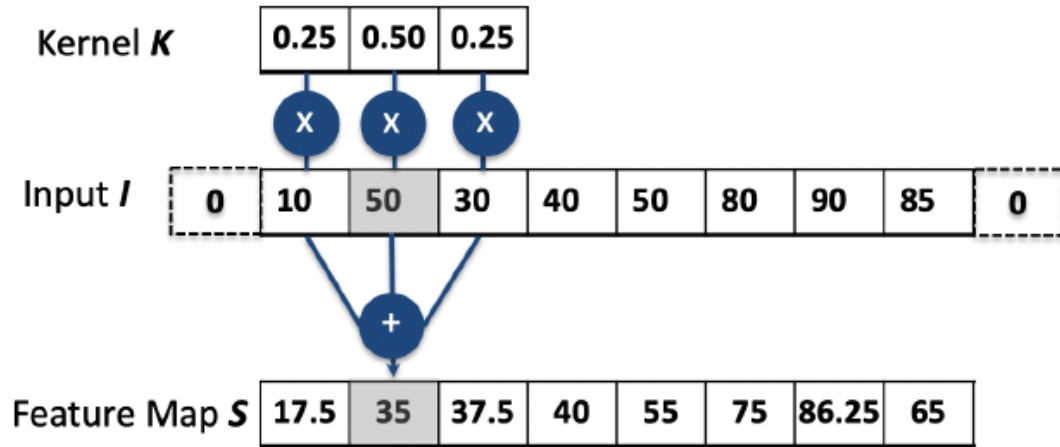


Module 8. Deep Learning

- ❑ Motivation
- ❑ Improve Training of Deep Learning Models
- ❑ Convolutional Neural Networks 
- ❑ Recurrent Neural Networks
- ❑ Summary

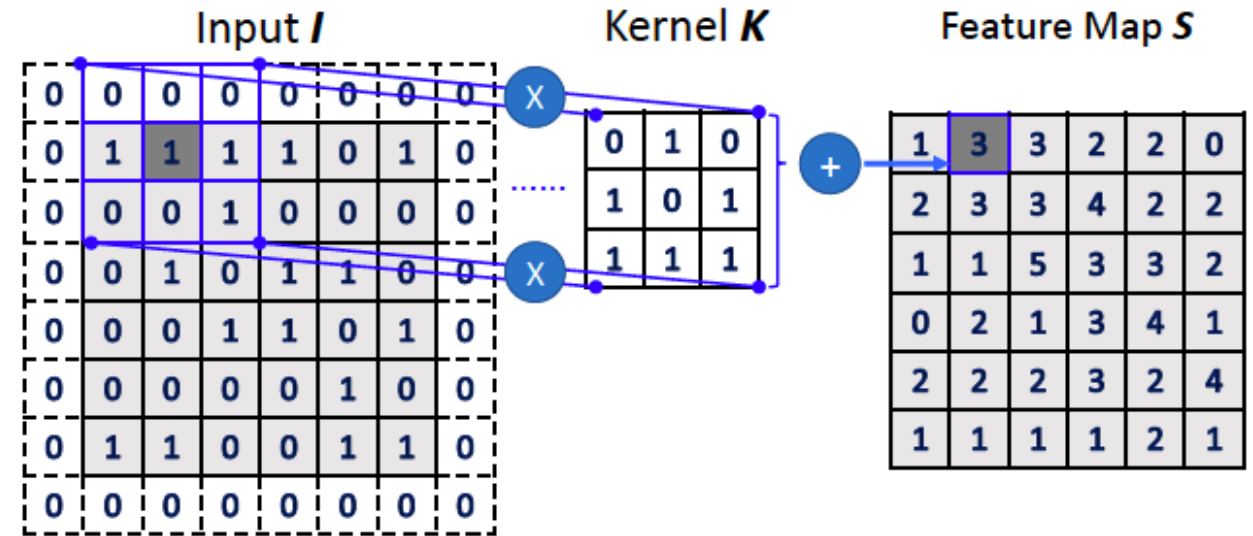
Convolutional Neural Networks (CNN)

- What is convolution? $S = I * K$



1D Convolution

$$S[i] = \sum_{p=-\frac{P-1}{2}}^{\frac{P-1}{2}} I[i+p]K[p], \quad (i = 1, \dots, N)$$

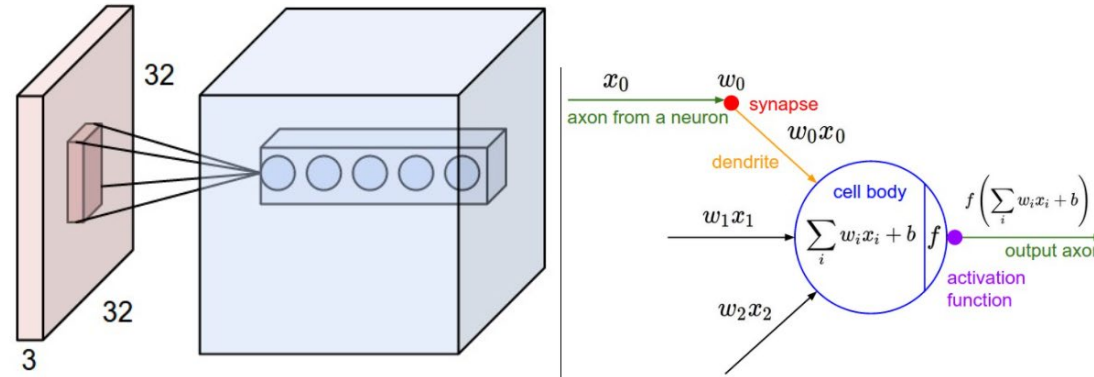


2D Convolution

$$S[i][j] = \sum_{q=-\frac{Q-1}{2}}^{\frac{Q-1}{2}} \sum_{p=-\frac{P-1}{2}}^{\frac{P-1}{2}} I[i+p][j+q] K[p][q]$$

- ❑ The outputs are computed by sliding a **kernel** (of weights) on the inputs, and computing weighted sum locally

A Convolutional Layer Example



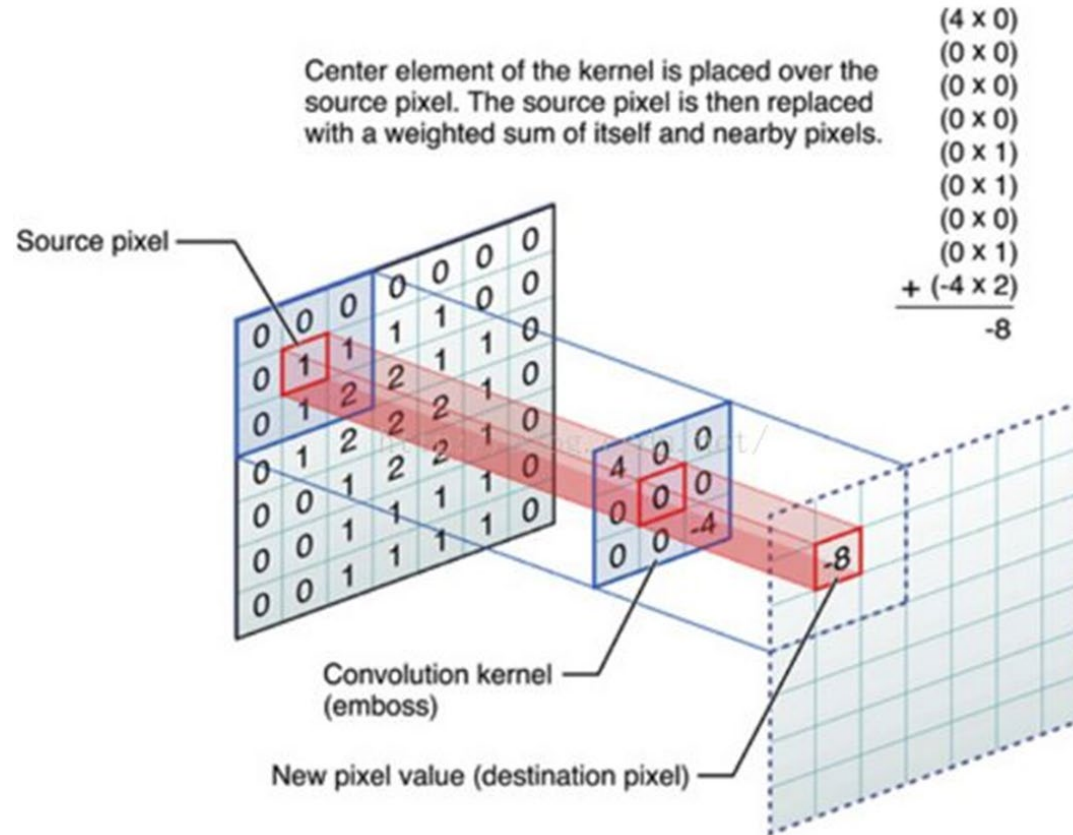
Left:

- ❑ An example input volume in red (e.g. a 32x32x3 CIFAR-10 image), and an example volume of neurons in the first Convolutional layer.
- ❑ Each neuron in the convolutional layer is connected only to a local region in the input volume spatially, but to the full depth (i.e. all color channels). Note, there are multiple neurons (5 in this example) along the depth, all looking at the same region in the input: the lines that connect this column of 5 neurons do not represent the weights (i.e. these 5 neurons do not share the same weights, but they are associated with 5 different filters), they just indicate that these neurons are connected to or looking at the same receptive field or region of the input volume, i.e. they share the same receptive field but not the same weights.

Right:

- ❑ The neurons from the Neural Network chapter remain unchanged: They still compute a dot product of their weights with the input followed by a non-linearity, but their connectivity is now restricted to be local spatially.

Convolutional Kernel



- Padding on the input volume with zeros in such way that the convolution layer does not alter the spatial dimensions of the input

Convolution Spatial Arrangement

- ❑ Three hyperparameters control the size of the output volume of the convolutional layer: the **depth**, **stride**, and **padding size**:
 - ❑ The **depth** of the output volume controls the number of neurons in a layer that connect to the same region of the input volume.
 - ❑ **Stride** controls how depth columns around the width and height are allocated. If the stride is 1, then we move the filters one pixel at a time. This leads to heavily overlapping receptive fields between the columns, and to large output volumes. However, in practice, $S \geq 3$ is rare. A greater stride means smaller overlap of receptive fields and smaller spatial dimensions of the output volume.
 - ❑ Sometimes, it is convenient to **pad** the input with zeros (or other values, such as the average of the region) on the border of the input volume. The size of this padding is a third hyperparameter. Padding provides control of the output volume's spatial size. In particular, sometimes it is desirable to exactly preserve the spatial size of the input volume, this is commonly referred to as "same" padding.
 - ❑ The spatial size of the output volume is $\frac{W-K+2P}{S} + 1$. If this number is not an integer, then the strides are incorrect and the neurons cannot be tiled to fit across the input volume in a symmetric way.